Heart Disease Prediction

Machine Learning Capstone Project

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Google Colab Notebook:

https://colab.research.google.com/drive/1Cw_rOmiSAVdnzCaoe3Beoxj8Lfy6A8vn?usp=sharing

GitHub Repository:

https://github.com/AyaKhaled22/ML_Project

I. Introduction

Project Goal

The objective of this project is to apply a complete Machine Learning pipeline to the UCI Heart Disease dataset. Three distinct tasks are addressed in this report: (1) Classification to predict the presence or absence of heart disease, (2) Regression to predict a patient's age from clinical features, and (3) Clustering to discover natural patient groups without using the diagnosis label.

Data Description

Source: UCI Machine Learning Repository (Heart Disease Dataset, ID = 45). Size: 303 rows x 14 columns. The dataset contains 13 clinical features and one target column (num) representing the diagnosis of heart disease with 5 severity levels (0-4).

Feature List:

- age: Age in years
- sex: Sex (1 = male; 0 = female)
- cp: Chest pain type (1-4)
- trestbps: Resting blood pressure (mm Hg)
- chol: Serum cholesterol (mg/dl)
- fbs: Fasting blood sugar > 120 mg/dl (1 = true; 0 = false)
- restecg: Resting electrocardiographic results (0-2)
- thalach: Maximum heart rate achieved
- exang: Exercise induced angina (1 = yes; 0 = no)
- oldpeak: ST depression induced by exercise
- slope: Slope of the peak exercise ST segment (1-3)
- ca: Number of major vessels colored by fluoroscopy (0-3)
- thal: Thalassemia (3 = normal; 6 = fixed defect; 7 = reversible defect)
- num: Diagnosis of heart disease (0 = no disease; 1-4 = disease severity)

II. Data Preprocessing & EDA

Data Cleaning

- Missing Values: Detected in 'ca' (4 missing, 1.32%) and 'thal' (2 missing, 0.66%).
- Strategy: Imputed using the mode (most frequent value) for both categorical features.
- Data Types: 'ca' and 'thal' were cast from float to int after imputation.
- Duplicate Rows: 0 duplicates found.

Feature Encoding

One-Hot Encoding was applied to the nominal categorical features: 'cp', 'restecg', and 'thal'. Feature dimensionality increased from 13 to 20 columns. The target 'num' was converted into a binary classification label (0 = Healthy, 1 = Diseased).

Feature Scaling & Data Split

- StandardScaler was applied to all numerical features (crucial for K-Means and Logistic Regression).
- Train/Test Split: 80% training (242 samples) and 20% testing (61 samples).
- Stratified sampling was used for the classification task to maintain class distribution.

Key Findings from EDA

- Balanced Dataset: 54.1% Healthy (164) vs 45.9% Diseased (139).
- Top positive correlations: 'thal' (0.52), 'ca' (0.46), 'exang' (0.43), 'oldpeak' (0.42), 'cp' (0.41).
- Strongest negative correlation: 'thalach' (-0.42), meaning higher max heart rate is linked to lower risk.
- 'fbs' and 'chol' showed very weak correlations (< 0.1) with the diagnosis.

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[Image not found: results/features_vs_target.png]

III. Modeling and Results

3.1 Regression Task: Linear Regression (Target = Age)

Model: Linear Regression. Objective: predict patient age from clinical features (the 'age' column was excluded from features to avoid data leakage).

Metrics on the Test Set:

- MSE: 4.3121
- RMSE: 2.0765 years
- R2 Score: 0.0512

Interpretation: The R2 score of ~0.05 indicates the model explains only 5% of the variance in age. This is a meaningful negative result: age cannot be predicted from clinical measurements such as blood pressure or cholesterol because these features describe heart condition, not the patient's biological timeline. This confirms no data leakage exists.

3.2 Classification Task: Logistic Regression (Target = Heart Disease)

Model: Logistic Regression (max_iter=1000). Baseline comparison: Decision Tree (max_depth=4).

Metrics on the Test Set:

- Logistic Regression Accuracy: 88.52%
- Decision Tree Accuracy: 78.69%
- Precision (Diseased): 0.86
- Recall (Diseased): 0.89

Confusion Matrix Analysis: The model correctly identified 29 out of 33 healthy patients and 25 out of 28 diseased patients. Only 3 False Negatives occurred, which is the most critical error type in a medical context (missing a sick patient is more dangerous than a false alarm). The high recall (89%) indicates the model is well-suited for clinical screening.

Precision vs Recall Trade-off: In heart disease screening, minimizing False Negatives (high Recall) is prioritized because missing a diseased patient could be fatal. The current balance (Precision=0.86, Recall=0.89) is excellent.

3.3 Clustering Task: K-Means

Method: The Elbow Method suggested K in the range of 5-8. Silhouette Scores were computed for K=2 to 10, and K=8 achieved the highest score (0.1658).

- Chosen K: 8
- Silhouette Score: 0.1658

Interpretation: A silhouette score of 0.17 is moderate, indicating that while the clusters are distinguishable, significant overlap exists. This is expected in medical data, where patient profiles exist on a continuum rather

than in distinct discrete groups.

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IV. Conclusion

Best Model per Task

Classification (Logistic Regression): Achieved 88.52% accuracy, making it the most successful model in this project. Heart disease can be reliably predicted from routine clinical measurements.

Regression (Linear Regression): Achieved R2 = 0.05, which is a poor score. However, this is expected: age is not a function of the clinical measurements collected in this dataset.

Clustering (K-Means, K=8): Achieved Silhouette Score = 0.17, indicating moderate separation. Medical data typically has natural overlap between patient groups.

Challenges Encountered

- The ambiguous target column 'num' contained 5 classes (0-4) that required careful binarization.
- Missing values were limited but required a categorical-appropriate imputation (mode).
- The regression task exposed a key lesson: not every target is predictable, and understanding 'why' is more valuable than forcing a good score.

Future Work

- Try advanced models (Gradient Boosting, XGBoost) for the classification task.
- Apply PCA before K-Means to reduce noise and potentially improve the Silhouette Score.
- Perform hyperparameter tuning (GridSearchCV) on the Logistic Regression model.
- Explore feature engineering and test on additional heart disease datasets for generalization.

The complete source code is available on Google Colab:
https://colab.research.google.com/drive/1Cw_rOmiSAVdnzCaoe3Beoxj8Lfy6A8vn?usp=sharing

And on GitHub: https://github.com/AyaKhaled22/ML_Project